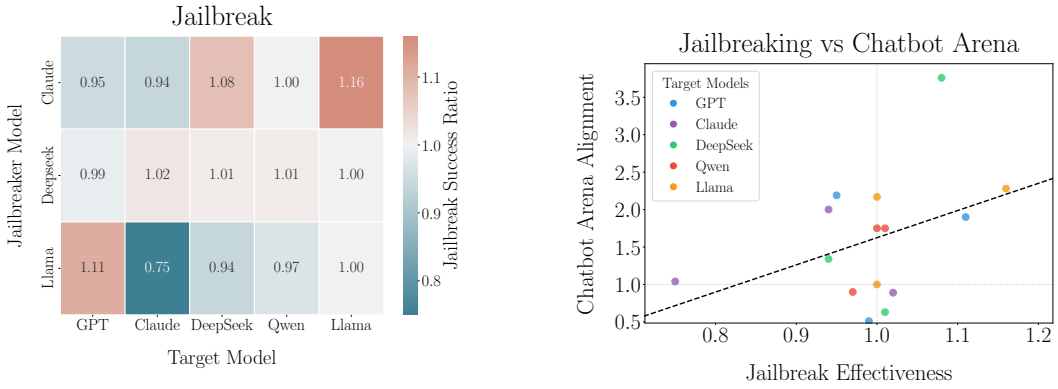


(a) (b)

Figure 7: **(a)** Ratio of identity-aware jailbreaking success versus identity-unaware. Each cell (i, j) represents the ratio of successful jailbreaks by jailbreaker model i against target model j when the jailbreaker is aware of the target's identity, relative to when it is unaware; **(b)** Scatter plot correlating models' strategic adaptation in preference alignment with their effectiveness to identity-aware jailbreaking. The y-axis represents the model's average alignment ratio from the Chatbot Arena experiment (Figure 6) when the model is aware of the judge or not. The x-axis represents the average jailbreaking success ratio when the target identity is revealed versus when it is not.

judge's identity to a player model leads to reward hacking. For each question, a player LLM generates two responses: one while knowing the judge's identity ('identity-aware') and one without this knowledge ('identity-unaware'). A judge LLM then performs a pairwise comparison, choosing the preferred response. To measure the effect of identity awareness, we compute a win-lose ratio: the number of wins for identity-aware responses divided by the number of wins for the identity-unaware responses. A ratio above 1.0 indicates that knowledge of the judge's identity helped the player model generate more preferred responses.

Results Figure 6 indicates that most models strategically adapt their responses when they are aware of the judge's identity. The majority of values in the win-lose ratio matrix are greater than 1.0, signifying that models generally perform better when informed about who will be judging their output. A few examples to illustrate the behavior adaptation are shown in Table 9 and Table 10. This finding lends support to the occurrence of reward hacking, as models appear to optimize their responses for specific known judges. DeepSeek stands as a notable exception, demonstrating decreased performance (0.63) when it knows it will judge itself and generally showing less adaptation to known judge identities across most conditions. This unique behavior warrants further investigation. Judge-specific patterns are also evident. DeepSeek appears to be the most challenging judge to adapt to. In contrast, Llama, as a judge, elicits strong adaptation from most generator models, which may be due to the difference in their release dates. Overall, current models tend to exhibit an ability to adapt to a known judge strategically, which raises concerns that models may superficially satisfy recognized evaluators while maintaining underlying misalignment, creating false confidence in safety assessments. We include additional qualitative results and discussions in Appendix F.

7 Case Study 3: Safety Threat

> *"Know your enemy and know yourself; in a hundred battles, you will never be in peril."*
>
> – The Art of War (Sun Tzu, 1963)

Motivation An LLM's identity can also be associated with its weakness. With the disclosure of the interacting agent's identity, an LLM can leverage its knowledge to exploit the interacting agent and cause safety threats among LLM interactions. Such risks can be profound when companies and individuals increasingly delegate responsibilities to LLM agents.

Setup We implement a controlled experimental design with two agent roles: jailbreaker models attempting to elicit prohibited content, and target models configured to resist such attempts. Following a methodology parallel to our Chatbot Arena experiments (Chiang et al., 2024) in Section 6, we systematically vary the information provided to jailbreaker models. In the experimental condition, we

explicitly reveal the target model's identity to the jailbreaker before prompting it to reframe a harmful query from the JailbreakBench dataset (Chao et al., 2024). In the control condition, the jailbreaker attempts to accomplish the same objective without knowing the target's identity. We quantify the effectiveness of jailbreaking using the success ratio metric, defined as the proportion of successful jailbreaks achieved with identity-aware prompts relative to those achieved with identity-unaware baseline prompts for each target model.

Results Figure 7a shows an insignificant pattern of increased success based solely on the jailbreaker's awareness of the target's identity. This can be attributed to the strong safety capabilities of the examined models. However, a more nuanced relationship emerges when we correlate these jailbreaking outcomes with the models' adaptive behaviors observed in the preference alignment experiments. Figure 7b presents a scatter plot correlating a model's tendency to adapt to known judges in the Chatbot Arena with its success ratio in identity-aware jailbreaking. We observe a moderate positive linear trend, with a Pearson correlation coefficient of $r = 0.394$. This correlation suggests that models exhibiting a greater capacity for strategic adaptation in preference alignment (i.e., they are better at "reward hacking" or aligning with a known judge's preferences) also tend to be more successful in jailbreaking when their targets' identities are revealed. In essence, a jailbreaker that can effectively map a target's identity to its likely response patterns and alignment characteristics can be better equipped to craft successful jailbreak prompts. Qualitative examples are shown in Appendix G.

8 Related Work

LLM situational awareness Situational awareness for AI models has recently emerged as a key concern in recent AI-safety research. It was first introduced by Cotra (2021) and formally discussed in Ngo et al. (2022) and Anwar et al. (2024). The term commonly refers to the capability of AI models to make decisions based on abstract knowledge about themselves and their situation. Berglund et al. (2023) leverages out-of-context reasoning to demonstrate the emergence of situational awareness in LLMs. More comprehensive tests have been created to examine LLM's capacity to recognize their own generated text and predict their behaviors (Laine et al., 2024). Recent studies have extended the exploration to investigate awareness of the environment and the future (Tang et al., 2024), as well as the user's preferences inferred from implicit cues (Jin et al., 2024a). While these efforts gain insights on how models understand themselves and their surroundings, they leave unexplored whether LLMs can perceive the identities of their interacting partners, which is explored in our study.

Multi-LLM systems Multi-LLM systems open new avenues for studying interactions among autonomous agents with natural language communication. Researchers have deployed such systems to simulate economic markets (Li et al., 2023), strategic gameplay (Xu et al., 2023), and world wars (Hua et al., 2023), demonstrating diverse emergent behaviors. The communication among agents has been harnessed for collaborative problem-solving (Zhang et al., 2024) and collaboration through debate (Xiong et al., 2023). Evaluating the diverse abilities of LLMs through their interactions is of particular interest. Piatti et al. (2024) examine LLMs' capability of long-term planning through sustainable fishing. Multi-turn negotiation among LLM agents is also explored to evaluate reasoning under conflicting objectives (Davidson et al., 2024; Xia et al., 2024). In this work, we build on previous efforts leveraging direct multi-LLM interactions to investigate interlocutor awareness.

9 Conclusion

Our study provides the first systematic evaluation of interlocutor awareness of LLMs. It shows evidence that LLMs can discern their interlocutors' model family, with better performance for in-family recognition and some ability to detect certain out-of-family models via cues or conversation. Our further demonstrations through case studies indicate that awareness of an interlocutor's identity can prompt behavioral adaptations, such as adjusting to a collaborator's capabilities, aligning with known judges, and a trait that correlates with increased vulnerability to identity-aware jailbreaking. These findings suggest both opportunities and risks: while interlocutor awareness might enable nuanced collaboration, it also introduces potential challenges to evaluation fairness, model security, and ethical AI interactions. Our work serves as the first step in raising the awareness of LLMs among interlocutors. It opens up new research directions on the question posed by our results: whether LLMs should retain their individual characteristics or be standardized

to avoid identity inference. We hope that this study will inspire further research and discussions on the applications and risks of interlocutor awareness.

Limitations

While our study provides novel insights into LLM interlocutor awareness and behavioral adaptation, several limitations warrant acknowledgment and suggest directions for future research.

Definition of model identity: For simplicity and experimental control, this study primarily defined an LLM's identity by its model family. However, "identity" can be a multifaceted concept, encompassing model size, specific versions, fine-tuning adaptations, or even personas adopted by the model. Investigating these more granular aspects of identity could provide a more nuanced understanding of how LLMs perceive and react to each other.

Prompt design and coverage: The prompts used for eliciting responses and for the specific tasks (e.g., identification prompts, conversation starters, harmful question reframing) were standardized for consistency. We predominantly used a single core prompt structure for each experimental condition. There could be potentially unintended biases introduced by a specific prompt template. Future work should iterate over semantically equivalent but structurally different prompts to mitigate any template-induced bias.

Data sampling and scale: Due to computational and API cost considerations, our experiments were conducted on randomly sampled subsets of 100 datapoints from each dataset. While this provides indicative results, larger-scale experiments across the full datasets could offer more statistically robust findings and potentially uncover less frequent but significant interaction patterns.

Ethical Considerations

Our research on interlocutor identity awareness raises important ethical considerations with dual-use implications. While interlocutor awareness can enhance collaborative capabilities and enable more effective human-AI interactions, our findings reveal potential vulnerabilities in evaluation frameworks and safety measures. By documenting these phenomena, we aim to inform more robust alignment techniques and evaluation protocols. LLMs' ability to strategically adapt to known evaluators threatens the integrity of systems like Chatbot Arena. This reward hacking undermines objective assessment of model performance, potentially creating misleading impressions of progress. Developers should implement safeguards such as anonymizing model identities during evaluations. Our jailbreaking experiments demonstrate how interlocutor awareness could compromise safety guardrails. We used controlled harmful prompts from established benchmarks but acknowledge potential risks.

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